

Deep Learning Image Classification Using Transfer Learning

Introduction

Deep learning has become one of the most powerful approaches for solving image classification problems. Convolutional Neural Networks (CNNs) are widely used in computer vision tasks because they can automatically learn image features and patterns. In this project, a deep learning image classifier was developed using transfer learning and a pretrained CNN model to classify images from a subset of the CIFAR-10 dataset.

The main objective of the project was to build an accurate image classification system while reducing training time through the use of pretrained models.

Dataset Description

The project used a subset of the CIFAR-10 dataset, which contains labeled images from different object categories such as airplanes, cars, birds, cats, dogs, ships, and trucks. The dataset consists of small RGB images commonly used for image classification research and experimentation.

The dataset was divided into:

- Training dataset
 - Validation dataset
 - Testing dataset
-

Data Preprocessing and Augmentation

Before training the model, several preprocessing steps were applied:

- Images were resized to match the input size required by the pretrained model.
- Pixel values were normalized to improve training stability.
- Data augmentation techniques were applied to improve model generalization and reduce overfitting.

The augmentation methods included:

- Random horizontal flipping
- Rotation
- Zooming
- Image shifting

These techniques helped generate more diverse training samples.

Deep Learning Model

Transfer learning was implemented using a pretrained CNN architecture such as MobileNetV2 or ResNet50. The pretrained layers were used as feature extractors, while custom fully connected layers were added for classification.

The model was trained using:

- Adam optimizer
- Categorical cross-entropy loss
- Batch training
- Multiple epochs

Initially, the pretrained layers were frozen, and only the custom layers were trained. Later, some pretrained layers were fine-tuned to improve performance.

Model Evaluation

The trained model was evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Training and validation accuracy/loss curves were plotted to monitor the learning process and detect overfitting.

The model achieved strong validation accuracy and demonstrated effective classification performance on unseen images.

Results

The transfer learning approach significantly improved classification performance compared to training a CNN from scratch. Data augmentation further improved the model's ability to generalize to new images.

The final model achieved:

- High classification accuracy
- Reduced overfitting
- Faster training time

The saved trained model can be reused later for inference without retraining.

Conclusion

This project successfully demonstrated the implementation of a deep learning image classification system using transfer learning. By combining pretrained CNN models with data augmentation techniques, the model achieved reliable performance on image classification tasks.

The project highlights the importance of preprocessing, transfer learning, and evaluation metrics in deep learning applications. Future improvements could include hyperparameter tuning, larger datasets, and deployment of the model as a web-based application.